

Business Requirements Document

E-Commerce Customer Behavior & RFM Segmentation

1. Document Information

Field	Details
Project Title	E-Commerce Customer Behavior & RFM Segmentation
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Role	Business Analyst / Data Analyst
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Version	1.0
Status	Final

2. Business Background

The e-commerce business has accumulated customer and order data across multiple cities over a three-year period. However, the business does not have a structured approach to identify its most valuable customers, customers who are becoming inactive, or customers who have already been lost.

Without customer segmentation, marketing and retention activities may be applied uniformly across the customer base instead of being targeted according to customer behavior and value.

This project analyzes customer purchasing behavior using historical order data and applies RFM (Recency, Frequency, Monetary) analysis to create meaningful customer segments. The analysis also examines revenue performance across product categories, cities, and membership tiers to support business decision-making.

3. Business Objective

The objective of this project is to analyze customer purchasing behavior and segment customers using RFM methodology.

The analysis aims to:

- Identify high-value and loyal customers.
- Identify customers who are at risk of becoming inactive or lost.
- Understand revenue contribution across different customer segments.
- Analyze revenue patterns by product category, city, and membership tier.
- Provide an interactive dashboard that enables stakeholders to explore customer and revenue trends.
- Provide data-driven recommendations for customer retention and membership growth.

4. Stakeholders

Stakeholder	Role	Requirement / Interest
Marketing Manager	Primary	Identify customer segments for targeted retention and marketing campaigns.
Product Manager	Secondary	Understand product category and membership tier performance.
Sales Team	Secondary	Identify high-value and at-risk customers for prioritized outreach.
Data Analyst	Internal	Perform data preparation, RFM calculations, analysis, and reporting.
Business Owner / Leadership	Approver	Understand overall customer health, revenue concentration, and business opportunities.

5. Scope

The following activities are included within the scope of this project:

- Analysis of historical customer and order data.
- Calculation of Recency, Frequency, and Monetary values.
- RFM scoring and customer segmentation.
- Identification of customer segments such as Champions, Loyal Customers, Potential Loyalists, At Risk, Need Attention, and Lost Customers.
- Revenue analysis by product category.
- Revenue analysis by city.
- Revenue analysis by membership tier.
- Calculation of supporting KPIs such as Average Order Value and Return Rate.
- Development of an interactive dashboard for business stakeholders.
- Development of data-driven business recommendations.

5.1 Out of Scope

The following activities are outside the scope of this project:

- Real-time data processing or real-time dashboard updates.
- Automated marketing campaigns or email delivery.
- Individual customer-level marketing automation.
- Predictive customer churn modeling.
- Machine learning-based customer prediction.
- Competitor analysis.
- Inventory management analysis.
- Development of a production e-commerce application.

5.2 Assumptions

- The available customer and order datasets are assumed to be representative of the business activity being analyzed.
 - The analysis covers the historical period available in the dataset.
 - RFM scoring thresholds are based on the methodology defined for this project.
 - Revenue calculations are based on delivered orders as defined in the analysis.
 - The dashboard is intended for analytical and decision-support purposes rather than automated operational execution.
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6. Functional Requirements

ID	Functional Requirement	Priority
FR01	The analysis shall calculate Recency, Frequency, and Monetary values for each customer.	High
FR02	The analysis shall assign customers to defined RFM-based customer segments.	High
FR03	The dashboard shall display customer distribution across RFM segments.	High
FR04	The dashboard shall display revenue by product category.	High
FR05	The dashboard shall display revenue performance by city.	Medium
FR06	The dashboard shall allow analysis of customer and revenue performance by membership tier.	Medium
FR07	The analysis shall calculate supporting KPIs including Average Order Value and Return Rate.	Medium
FR08	The dashboard shall provide interactive filtering to allow stakeholders to explore the analysis.	High
FR09	The analysis shall identify revenue contribution from different customer segments.	High
FR10	The project shall provide business recommendations based on the analytical findings.	High

7. Non-Functional Requirements

- The dashboard shall be understandable to non-technical business stakeholders.
- The analysis shall use consistent definitions and calculations across all reports and visualizations.
- Dashboard visualizations shall use clear labels and consistent formatting.
- The analysis should allow stakeholders to interpret customer and revenue trends without requiring knowledge of SQL.
- Project documentation and analytical outputs shall be maintained in a structured and version-controlled repository.
- The analysis should provide sufficient transparency for stakeholders to understand how customer segments and KPIs were derived.

8. Expected Business Outcome

The completed analysis is expected to provide the business with better visibility into customer behavior and revenue patterns.

The resulting customer segmentation can help stakeholders distinguish between high-value customers, loyal customers, customers requiring attention, and customers who may have already been lost.

The dashboard and recommendations provide a foundation for more targeted customer retention, marketing prioritization, and membership strategy decisions.
